

Quantization for Efficient LLM Inference: A Quality–Memory–Speed Study of Qwen2.5-7B-Instruct

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Abstract

Quantization lets large language models run on cheaper hardware, but a smaller model is not automatically a better one: it can lose accuracy, run slower in some runtimes, or save disk without saving the memory that actually limits inference. We measure these trade-offs for one 7B-class instruction model, Qwen2.5-7B-Instruct. Against a bf16 baseline we compare 8-bit and 4-bit post-training quantization (LLM.int8(), NF4, and official GPTQ and AWQ) on capability benchmarks (MMLU, GSM8K, HumanEval, IFEval) and a set of efficiency measurements on a single RTX 4090. The 4-bit GPTQ and AWQ checkpoints give the best overall trade-off: they are the only variants that cut both disk and GPU memory while staying close to baseline speed, at a modest quality cost. Two results run against intuition: LLM.int8() is the slowest variant of all and saves no disk, and NF4 saves memory but still decodes slower than bf16. Together they show that quantization is mainly a memory technique rather than a speed one. We also implement and run four catalogued extensions on the same box: a JudgeBench meta-evaluation, W8A8/SmoothQuant self-quantization, a quantized KV cache, and QLoRA recovery. These sharpen the memory-versus-speed picture.

1 Introduction and Research Questions

Many LLMs are too expensive to serve in full precision, and post-training quantization is the standard remedy. Several practical methods exist: LLM.int8() [3] for 8-bit weights with mixed-precision handling of outliers; GPTQ [5] and AWQ [10], which use a small calibration set to push weights to 4 bits; SmoothQuant [15] for 8-bit weights and activations; the NF4 data type from QLoRA [4] for 4-bit storage in the bitsandbytes ecosystem; and KV-cache quantization such as KIVI [11] for long-context inference. None of these is automatically beneficial: the same 4-bit model can be smaller yet slower, or degrade unevenly across capabilities, so the practical trade-off is what we set out to measure. We organise the study around four questions from our proposal:

RQ1 Which method gives the best quality–memory–speed trade-off for a 7B-class instruction LLM?

RQ2 Are some capabilities (reasoning, coding, QA, instruction following) more sensitive to quantization than others?

RQ3 Does quantization improve real inference speed, or mainly reduce memory?

RQ4 (optional) Can light QLoRA-style adaptation recover quality lost after 4-bit quantization?

Our core study is the proposal’s minimal successful version (one model, five variants, four benchmarks, full efficiency profiling), and we then implement the catalogued extensions as a gated code layer and run the four that fit our GPU budget.

2 Experimental Setup

Model and variants. All experiments use Qwen2.5-7B-Instruct [13] at a pinned commit, run on a single on-demand RTX 4090 (24 GB, sm_89). Every variant, including the bf16 baseline, runs on the same GPU, so the speed comparison is apples-to-apples with no cross-hardware caveat. Table 1 lists the five core variants. LLM.int8() and NF4 quantize the baseline weights at load time, so their disk footprint is the full bf16 checkpoint; GPTQ and AWQ are official pre-quantized 4-bit checkpoints served by the gptqmodel Marlin kernels through optimum.

Benchmarks and protocol. Quality is measured with the EleutherAI lm-evaluation-harness [6]: MMLU [7] (accuracy, aggregated with a weighting over its 57 subjects), GSM8K [2] (exact-match under flexible extraction, since the strict-match number is a formatting artefact for instruct models), HumanEval [1] (pass@1, with code execution), and IFEval [16] (prompt-level strict accuracy). Decoding is greedy and deterministic, with `do_sample=False` set at both load and generation. MMLU and GSM8K use the chat template with 5-shot multi-turn prompting; HumanEval and IFEval are 0-shot. Efficiency is profiled separately (§3.2): the model is loaded once, then we measure disk size, load and generate peak VRAM, prefill and decode throughput at short, medium, and long prompts (128, 1024, and 4096 tokens), and the largest batch size before running out of memory.

Variant	Method	Precision	Source
bf16	half-precision baseline	W16	reference
bnb-int8	LLM.int8() [3]	W8 (load-time)	bitsandbytes
bnb-nf4	NF4 + double-quant [4]	W4 (load-time)	bitsandbytes
gptq-official	GPTQ [5]	W4 (g128)	official Qwen ckpt
awq-official	AWQ [10]	W4 (g128)	official Qwen ckpt

Table 1: The five core variants. Load-time means the disk checkpoint is the full bf16 model, quantized in memory during loading.

3 Results

3.1 Quality: capability sensitivity (RQ2)

Table 2 and Figure 1 report quality across the four benchmarks. The main finding is that quantization is remarkably benign for quality at 7B: every 4-bit variant stays within about 1.5 points of the bf16 baseline on MMLU and GSM8K. Sensitivity is uneven across capabilities, which answers RQ2.

Knowledge and math are close to lossless: MMLU and GSM8K barely move, and 8-bit is essentially unchanged. The MMLU shifts are small but, at a ± 0.4 standard error, resolvable; most other deltas are within roughly one standard error and should be read as trends rather than firm rankings. Coding shows the widest spread, with bnb-int8 losing about four points on HumanEval, the single largest drop; the 4-bit methods give up a point or two, though at HumanEval’s ± 3 -point standard error only the bnb-int8 gap approaches significance. Instruction following degrades only mildly, within about one standard error, and bnb-nf4 essentially ties the baseline. The 8-bit variant is the clearest illustration that more bits do not uniformly mean higher quality: it is near-lossless on knowledge and math yet gives the worst HumanEval score of any variant.

Variant	MMLU (acc)	GSM8K (flex. EM)	HumanEval (pass@1)	IFEval (strict)	Mean
bf16	73.6 ± 0.4	84.5 ± 1.0	84.1 ± 2.9	71.3 ± 1.9	78.4
bnb-int8	73.4 ± 0.4	84.4 ± 1.0	79.9 ± 3.1	69.3 ± 2.0	76.8
bnb-nf4	72.6 ± 0.4	83.6 ± 1.0	81.7 ± 3.0	71.0 ± 2.0	77.2
gptq-official	72.1 ± 0.4	83.2 ± 1.0	81.1 ± 3.1	70.2 ± 2.0	76.7
awq-official	72.2 ± 0.4	84.3 ± 1.0	81.1 ± 3.1	69.9 ± 2.0	76.9

Table 2: Quality (%), with standard error ($\pm$) shown per benchmark. MMLU is weighted over 57 subjects; GSM8K uses flexible-extract exact-match. Mean is a plain average across the four benchmarks (no standard error). Best per column in bold.

Quality by benchmark and quantization variant

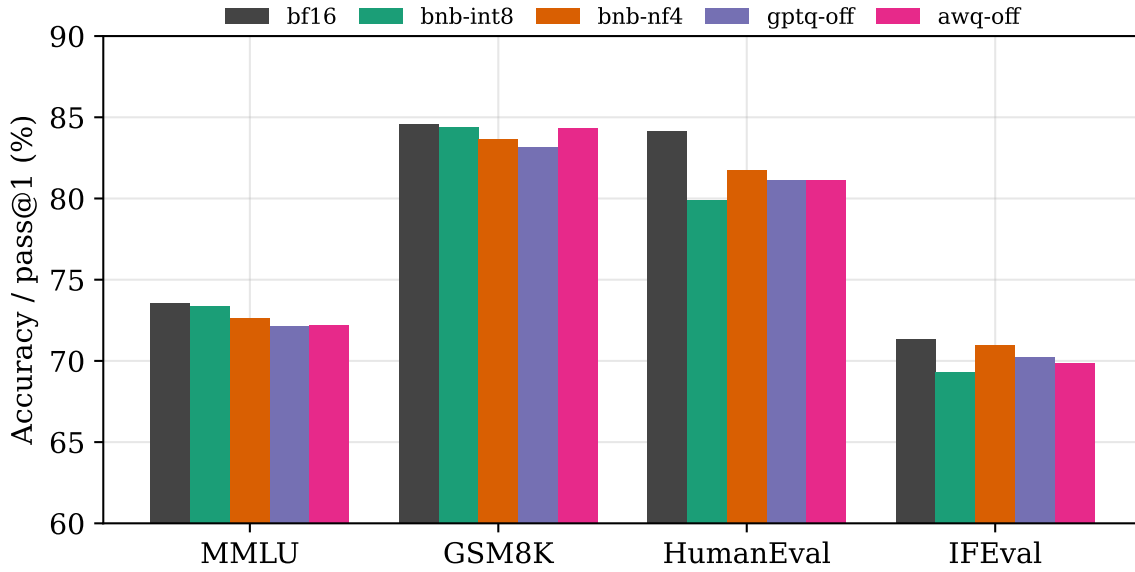


Figure 1: Quality by benchmark and variant. The 4-bit methods cluster tightly; HumanEval (coding) shows the widest spread, with bnb-int8 the most degraded.

3.2 Efficiency: memory vs. speed (RQ3)

The efficiency profile (Table 3, Figure 2) gives the clearest and most surprising findings, and answers RQ3 directly: quantization is primarily a memory technique, not a speed one.

LLM.int8() is the slowest variant. Its decode throughput collapses to about five times slower than bf16, because of the mixed-precision outlier decomposition, and it saves no disk since it stores the full bf16 checkpoint, making it the worst of both worlds on the speed and disk axes. NF4 is the memory-not-speed case: it cuts generate VRAM sharply and doubles the maximum batch, but still decodes below bf16. GPTQ and AWQ are the only variants that improve memory without paying much in speed. They cut disk by about 2.7 times, roughly halve generate VRAM, double the maximum batch, and hold decode to within about a quarter of baseline (28–29 vs. 37 tok/s) — a modest slowdown next to the 2–5× of the other quantized variants. Prefill is unaffected across all variants, because quantization touches the memory-bound decode phase rather than the compute-bound prefill.

Variant	Disk (GiB)	Gen. VRAM (GiB)	Max batch	Decode (tok/s)	Prefill (tok/s)
bf16	14.19	14.96	16	37.0	9505
bnb-int8	14.19	9.22	16	7.2	9220
bnb-nf4	14.19	5.95	32	16.9	9196
gptq-official	5.19	5.94	32	28.5	9699
awq-official	5.19	5.96	32	27.9	9615

Table 3: Efficiency on one RTX 4090. Disk is the on-disk weight bytes; max batch is the largest power-of-two before OOM at 2048-token sequences; decode and prefill are at the 4096-token (long) bucket. Per-bucket throughput is in the appendix (Table 4).

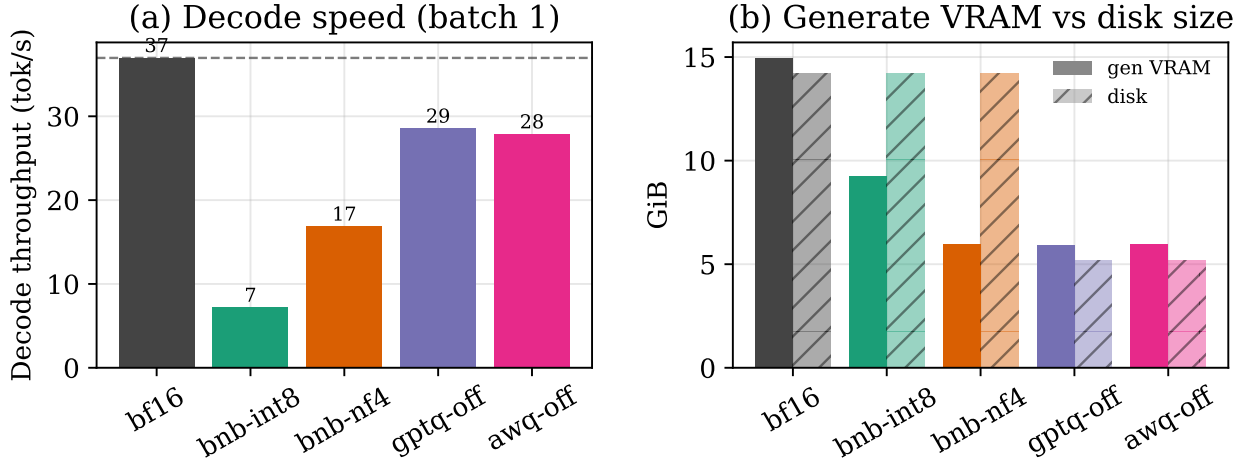


Figure 2: (a) Decode throughput; the dashed line marks bf16. bnb-int8 is about five times slower, and only GPTQ and AWQ stay near baseline. (b) Generate VRAM against on-disk size: the bitsandbytes variants shrink VRAM but not disk, while GPTQ and AWQ shrink both.

3.3 The trade-off and a recommendation (RQ1)

Figure 3 places all variants in the quality–speed–memory space, and the Pareto picture is clear. bf16 buys top speed at the highest memory; bnb-int8 is dominated, being both slow and disk-heavy; bnb-nf4 trades speed for memory; and GPTQ and AWQ sit at the sweet spot, with roughly half the memory, the largest batch, and decode within about 25% of baseline (a modest slowdown, versus 2–5× for the other methods) for a small quality cost. Under limited VRAM we recommend an official GPTQ or AWQ checkpoint. NF4 is the best fallback when no pre-quantized checkpoint exists, since it needs only the base weights, and LLM.int8() is hard to recommend on this hardware.

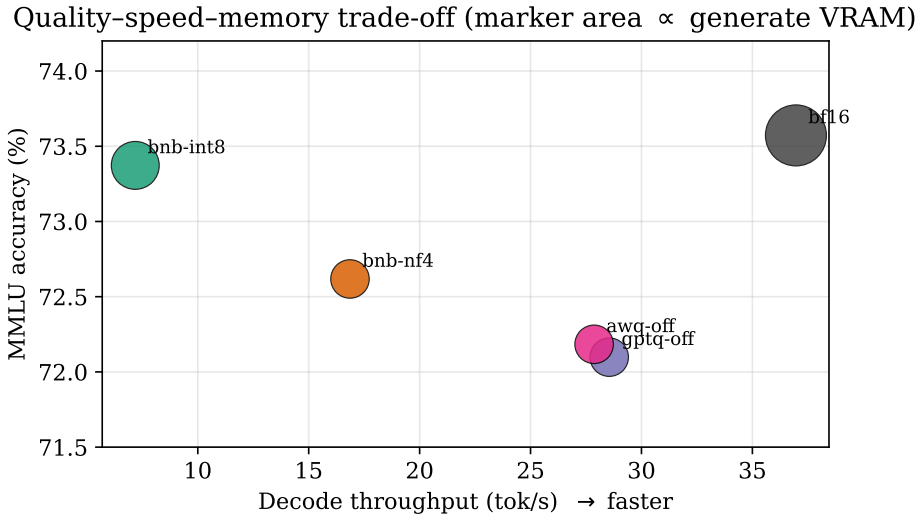


Figure 3: Quality–speed–memory trade-off. Up and to the right is better (higher MMLU, faster decode); smaller markers use less generate VRAM. GPTQ and AWQ occupy the high-speed, low-memory corner, and bnb-int8 is Pareto-dominated.

4 Extensions

We implemented all five catalogued extensions as a gated, separate code layer (`qquant.ext`) that reuses the core harness without touching it, and ran the four that fit the GPU budget; adding Mistral-7B as a second model was skipped because it would re-spend the entire compute budget. The whole extension run cost under \$1 of GPU time.

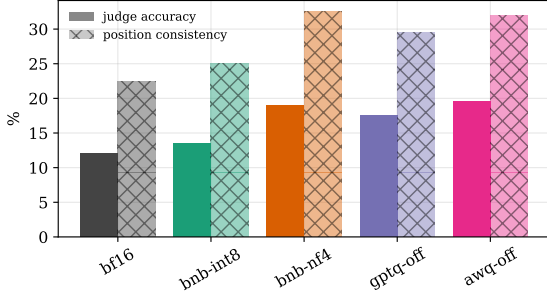
EXT-2: JudgeBench. We evaluate each variant as a pairwise judge on JudgeBench [14], a set of objective-correctness pairs, scoring agreement with the gold label under position-bias control, where both the A/B and B/A orders must agree. Figure 4a shows that quantization does not degrade judging: the 4-bit variants are, if anything, marginally better than bf16. The absolute numbers are low because Qwen2.5-7B is a weak, heavily position-biased judge on these hard pairs (only 22 to 33% of its verdicts are order-consistent), a property of the base model rather than of quantization.

EXT-3: W8A8 and SmoothQuant. We self-quantized a W8A8 checkpoint [15] (per-channel int8 weights with int8 activations) from the same C4 calibration set as our 4-bit self-quant recipes, using llm-compressor [12]. The checkpoint is produced and loads natively, but two things emerged on the HF runtime. First, compressed-tensors decompresses W8A8 to a bf16-resident tensor (15.2 GiB in VRAM), so it yields disk-only savings and no inference-memory benefit. Second, its `generate()` path is severely runtime-confounded on sm_89, running at about 4.5 s per iteration, so a full quality sweep is impractical: the slowdown is itself the result. W8A8 is best understood as a vLLM or kernel-path technique, not an HF-generate one.

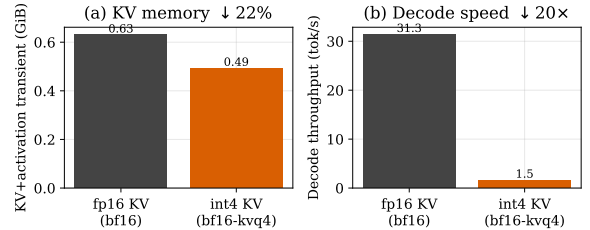
EXT-4: Quantized KV cache. Using the transformers v5 quantized cache (quanto, int4), we measured peak generation VRAM at a 3072-token context (Figure 4b). The int4 KV cache trims the KV-plus-activation transient by about 22%, but slows decode by roughly 20 times, because quanto dequantizes the cache on every step. This is a clear, quantitative confirmation of the runtime-confounded throughput caveat: a quantized KV cache trades a modest memory saving for a large ($\sim 20\times$) speed cost on the HF backend, and is only worthwhile with a fused kernel path such as KIVI [11] or vLLM.

EXT-5: QLoRA recovery (RQ4). We verified the full QLoRA [4, 8] recovery pipeline end-to-end: a LoRA adapter of rank 16, trained on the frozen NF4 base over general instruction data and guarded by a fail-fast check that the training corpus never overlaps the evaluation benchmarks, together with an adapter-aware load path. Training converged, and the adapted model loads with the adapter active and generates coherently. A full recovery measurement (enough training steps to move benchmark accuracy, followed by a re-evaluation) is a longer campaign we leave to future work; here we contribute the validated, contamination-safe machinery.

EXT-2 JudgeBench: pairwise judge accuracy & position consistency



(a) EXT-2 JudgeBench



(b) EXT-4 int4 KV cache

Figure 4: Extension results. (a) Judge accuracy and position-consistency per variant: quantization does not hurt judging. (b) int4 KV cache: about 22% less memory transient but roughly 20 times slower decode.

5 Conclusion

For a 7B-class instruction model on a single RTX 4090, 4-bit GPTQ and AWQ are the best default under limited VRAM: they cut disk and generate VRAM by more than half, double the feasible batch size, and keep decode near baseline, all for a small quality cost (RQ1). Capability degradation is uneven, with coding the most sensitive and knowledge and math the least (RQ2). Most importantly, quantization is a memory technique rather than a speed one (RQ3): LLM.int8() is about five times slower than bf16 and saves no disk, NF4 saves memory while still decoding below baseline, and only the calibrated 4-bit kernels of GPTQ and AWQ deliver memory savings for only a modest ($\sim 20\text{--}25\%$) decode slowdown. The extensions reinforce this: W8A8 and the int4 KV cache both work yet are runtime-confounded on the HF backend, and quantization does not degrade judging in this small-sample test. The main limitations are single-seed evaluation and the

runtime confound on the self-quant and KV paths, which a fused-kernel serving stack such as vLLM would resolve; multi-seed runs, the second model (Mistral-7B [9]), and a full QLoRA recovery measurement (RQ4) are the natural next steps.

LLM Usage

In line with the course policy, we document our use of AI tools. The research design (the variant matrix, the quality–memory–speed framing, and the fairness and determinism protocol) is ours. We used an agentic coding assistant for supporting engineering: the torch-free orchestration layer and vast.ai run drivers, the evaluation and profiling harness scaffolding, the extension code layer, and the figure- and table-generation scripts. Every quantitative claim in this report was extracted programmatically from the committed result files and cross-checked, and agent-written code and prose were verified against the codebase and the logged metrics before inclusion.

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A Reproducibility and Selected Efficiency Details

Setup. Single on-demand RTX 4090 (24 GB, sm_89) on vast.ai; torch 2.12.1+cu126, transformers 5.10.1, lm-eval 0.4.3 (0.4.12 for HumanEval, whose code-execution harness requires the newer release), bitsandbytes 0.49.2, gptqmodel 7.1.0, compressed-tensors 0.17.1. Greedy decoding; seeds fixed (random, numpy, torch, few-shot). Quality and efficiency are profiled in separate phases, and each variant is loaded once. All result JSONs and the extension and figure code are in the project repository; the figures and tables in this report are generated directly from those files.

Variant	Bucket	Prompt (tok)	Prefill (tok/s)	Decode (tok/s)	TTFT (s)	E2E lat. (s)
bf16	short/long	128/4096	4141/9505	37.0/37.0	0.03/0.43	6.93/7.33
bnb-int8	short/long	128/4096	-/9220	7.2/7.2	—	—
bnb-nf4	short/long	128/4096	-/9196	16.9/16.9	—	—
gptq-official	short/long	128/4096	-/9699	28.3/28.5	—	—
awq-official	short/long	128/4096	-/9615	28.0/27.9	—	—

Table 4: Per-bucket throughput (selected fields; short/long = 128/4096-token prompts). Decode is near-constant across prompt length, as expected for a memory-bandwidth-bound phase. Full TTFT and latency per bucket are in the result JSONs.

Extension artifacts. EXT-2 produced one JudgeBench cell per variant (n=200 pairs, position-bias controlled). EXT-3 produced an 8.7 GiB W8A8 compressed-tensors checkpoint (per-channel int8 weights, int8 activations) from the shared C4 calibration set. EXT-4 measured the int4 quantized KV cache at a 3072-token context. EXT-5 produced a rank-16 LoRA adapter (general instruction data, contamination-guarded) and a verified adapter-attach load path.